

Subject Description Form

Subject Code	EEE505
Subject Title	Design and Simulation of Photonic Circuits
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students an understanding of the photonic integrated circuits (PICs). Fundamental design concepts of photonic devices will be introduced. The subject focuses on the design and simulation of photonic circuits. Design rules and layout will be studied. Packaging technologies and measurement of the photonic devices will also be introduced.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills - Understand the basic concept of photonic circuits. Gain knowledge of the basic operational principles of an optical waveguide, modulator, photodetector, resonator, etc. - Understand the design rule and layout, principles of PDK (process design kits), MPW (multi-project wafer services), and tape-outs. - Understand the design of the components, including splitters, resonators, and couplers, etc. Category B: Attributes for all-roundedness - Communicate effectively. - Think critically and creatively. - Assimilate new technological development in related field
Subject Synopsis/ Indicative Syllabus	Introduction of photonic integrated circuits (PICs): Understanding of basic concepts in photonic ICs, design flows, the principle of PDK. Introduction of circuit design: Learn the standard components of a design library and build a simple passive circuit. Overview of waveguides: Basic operational principle of an optical waveguide, and the impact of the material chosen on the performance of the waveguide. Components design and simulation: Component description and introduction to simulation techniques. Photonic circuits manufacturing and applications: Impact of different processes steps on components and circuits. Various applications of PICs.
Teaching/Learning Methodology	The basic concepts of photonic components will be introduced in lectures. The student will understand the PIC design flow and will be able to design and simulate basic photonic components. They will understand the nonidealities in photonic chips and circuits. Students will also be required to study one of the photonic devices and share their findings with other classmates through presentations.

	Teaching/Learning Methodology	Intended subject learning outcomes						
		a	b	c	d	E	f	
	Lectures	✓	✓	✓	✓	✓		
	Assignments & Exercises	✓	✓	✓	✓	✓	✓	
	Case study and report		✓	✓	✓	✓	✓	
	Examinations	✓	✓	✓				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Assignment	30%	✓	✓	✓	✓	✓	✓
	2. Simulation and report	20%	✓	✓	✓	✓	✓	✓
	3. Examination	50%	✓	✓	✓			
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	Assignments: Assignments are given to students to assess their competence level of knowledge and comprehension. The criteria (i.e., what to be demonstrated) and level (i.e., the extent) of achievement will be graded. These will be made known to the students before an assignment is given. Feedback about their performance will be given promptly to students to help them to improve their learning.							
	Simulation and reports: Students will be required to perform a simulation and submit a report, by reviewing the material taught in the lectures as well as their self-study and readings.							
	End-of-semester Examination: There will be an end-of-semester examination to assess students’ achievement of all the learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lectures/Tutorials					24 Hrs.		
	▪ Simulation and presentation					15 Hrs.		
	Other student study effort:							
	▪ Reading and self-study					60 Hrs.		
	▪ Assignment					21 Hrs.		
	Total student study effort					120 Hrs.		

Reading List and References	Textbooks • Chrostowski, L., and Hochberg, M. E. Silicon Photonics Design: From Devices to Systems, Cambridge University Press. 2015. • Reed, G. T. Silicon photonics: the state of the art. John Wiley & Sons, Ltd. 2008. References • L. Pavesi & D J Lockwoodt, Silicon Photonics. Springer, 2004.
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